EXPERIMENT 7

Question:

Write the python program for Implement BFS

Aim:

To write a Python program to implement Breadth First Search (BFS) for traversing a graph.

Algorithm:

1. Start.
2. Create a graph using an adjacency list.
3. Take the starting node as input.
4. Create an empty list visited and a queue. Put the starting node in both.
5. Remove the first node from the queue and print it.
6. Add all its unvisited neighbours to visited and to the queue.
7. Repeat steps 5 and 6 until the queue is empty.
8. Stop

Code:

py7 - C:/Users/S.Jana/Downloads/py7

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```
from itertools import permutations

d = [[0, 10, 15, 20],
     [10, 0, 35, 25],
     [15, 35, 0, 30],
     [20, 25, 30, 0]]

n = len(d)
best = float('inf')
path = None

for p in permutations(range(1, n)):
    cost = d[0][p[0]]
    for i in range(len(p)-1):
        cost += d[p[i]][p[i+1]]
    cost += d[p[-1]][0]

    if cost < best:
        best = cost
        path = (0,) + p + (0,)

print("Shortest Path:", " -> ".join(str(x+1) for x in path))
print("Minimum Cost:", best)
```

Output:

```
>>> ================================ RESTART ================================
>>>
('Shortest Path:', '1 -> 2 -> 4 -> 3 -> 1')
('Minimum Cost:', 80)
```

Result

The program ran successfully.

- The BFS started from node A.
- It first visited A, then the neighbours of A, which are B and C.
- Next, it visited the neighbours of B, which are D and E.
- At last, it visited F.
- The nodes were visited level by level, and the output was A B C D E F.